

**SMART CONNECT: ENHANCING
BRAND-INFLUENCER
COLLABORATIONS THROUGH AI**

A PROJECT REPORT

Submitted by

VENGADESAN G	311821205060
PRAJESH RAM V	311821205043
MANIKANDAN P	311821205023

*in partial fulfillment for the award of the degree
of*

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

Mohammed Sathak AJ College Of Engineering

Affiliated to Anna University , Chennai – 600 025

DECEMBER 2024

BONAFIDE CERTIFICATE

Certified that this project report **SMART CONNECT: ENHANCING BRAND-INFLUENCER COLLABORATIONS THROUGH AI** is the bonafide work of **VENGADESAN G., PRAJESH RAM V., MANIKANDAN P.**, who carried out the project work under my supervision.

HEAD OF THE DEPARTMENT

MRS. ANGAYARKANNI
Department of Information Technology
Mohammed Sathak AJ college of
Engineering
IT Sipcot , OMR , Siruseri ,
Chennai – 603103

SUPERVISOR

MRS. ANGAYARKANNI
Department of Information Technology
Mohammed Sathak AJ college of
Engineering
IT Sipcot , OMR , Siruseri ,
Chennai - 603103

Submitted for the project viva voce examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

This project introduces the development of a web-based platform designed to bridge the gap between companies and influencers, ensuring efficient and reliable collaboration for marketing campaigns. The system addresses common challenges in influencer marketing, such as ineffective matchmaking, lack of transparency, and bypassing platform protocols, by incorporating AI-driven algorithms for precise influencer recommendations based on campaign objectives, audience demographics, and engagement metrics.

The platform features a secure communication channel, automated contract generation, and integrated payment processing to foster trust and accountability. To further enhance reliability, the system includes performance tracking and analytics tools, empowering companies to assess campaign outcomes and influencers to refine their strategies.

Developed using React.js, Node.js, and Firebase, the web application incorporates a robust database for managing user profiles, campaign details, and transaction records. Its scalability ensures adaptability for businesses of various sizes and campaign complexities. This innovative solution redefines influencer marketing, promoting transparency, efficiency, and long-term collaboration between brands and influencers.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO
	ABSTRACT	iii
1.	INTRODUCTION	1
2.	LITERATURE REVIEW	2
3.	SYSTEM STUDY	4
4.	SYSTEM DESIGN	6
5.	SYSTEM REQUIREMENTS	7
6.	CONCLUSION	9

1. INTRODUCTION

Influencer marketing is essential for modern businesses, but traditional methods of collaboration often lack efficiency and transparency. Manual scouting or agency-based matchmaking is prone to errors, inefficiencies, and accountability issues.

To address these challenges, this project proposes a web-based platform that uses AI to connect companies with influencers based on metrics such as demographics, engagement rates, and campaign goals. The system incorporates features like secure communication, contract management, and performance tracking, ensuring transparency and reliability.

This innovative approach enhances the influencer marketing process, providing a robust and scalable solution for seamless brand-influencer collaborations.

2. LITERATURE REVIEW

EXISTING SYSTEMS

AGENCY-BASED INFLUENCER MARKETING

Traditional marketing agencies connect companies with influencers but often lack transparency and charge high fees. The process can be slow, with limited data-driven insights into campaign effectiveness.

SOCIAL MEDIA PLATFORMS

Platforms like Instagram and YouTube enable direct collaborations but lack dedicated matchmaking tools. Companies must manually search for influencers, leading to inefficiencies and potential mismatches.

INFLUENCER MARKETING PLATFORMS

Existing platforms provide basic matchmaking based on follower count and niche but often ignore deeper metrics like engagement quality or audience demographics. They also lack robust contract and payment management systems.

AI-POWERED TOOLS

Some platforms incorporate AI for influencer discovery but are limited in scope, focusing only on matching rather than managing the entire collaboration lifecycle.

Conclusion

While these systems address parts of the influencer marketing process, they fall short in offering a comprehensive, transparent, and data-driven solution, highlighting the need for an improved platform.

3. SYSTEM STUDY

PROPOSED SYSTEM

The proposed web-based platform introduces an innovative approach to influencer marketing by leveraging AI-driven matchmaking and secure collaboration tools. It addresses inefficiencies in traditional methods by providing a comprehensive system for companies and influencers to connect and collaborate effectively.

Core Mechanisms

1. AI-Driven Matchmaking:

- Companies post campaign requirements, including target demographics, goals, and budgets.
- The platform uses AI to analyze influencer profiles based on metrics like audience engagement, niche, and past campaign performance to recommend the most suitable influencers.

2. Secure Collaboration Tools:

- A built-in messaging system ensures all communications occur within the platform, preventing direct contact outside the app.
- Automated contract generation and secure payment gateways streamline the collaboration process, enhancing trust and accountability.

3. Performance Tracking:

- Companies can monitor campaign outcomes using detailed analytics, such as engagement rates, reach, and ROI.
- Influencers receive feedback on performance metrics, enabling them to refine their strategies for future campaigns.

Advantages of Dual-Entry Verification

- **Enhanced Efficiency:** Automates the process of influencer selection and campaign management, saving time and resources.
- **Transparency:** Maintains a digital audit trail of all interactions, contracts, and transactions, ensuring accountability.
- **Scalability:** The platform can handle diverse campaign sizes and requirements, catering to businesses of all scales.
- **Security:** Prevents circumvention through masked contact details, secure communication channels, and integrated legal frameworks.

Broader Implications

This platform has the potential to redefine influencer marketing by fostering fair, transparent, and efficient collaborations. Its emphasis on accountability and data-driven insights sets a benchmark for future advancements in the field.

In conclusion, the proposed system overcomes the limitations of existing methods by combining AI technology with innovative collaboration tools. It represents a significant step forward in creating a robust ecosystem for brands and influencers.

4 . SYSTEM DESIGN

TECHNOLOGY

STACK

ANGULAR

Angular is a popular front-end web application framework developed and maintained by Google. It is used for building dynamic, single-page applications (SPAs) with a focus on performance and user experience.

SPRINGBOOT

Spring Boot is a powerful Java-based framework designed for building and deploying microservices and enterprise-grade back-end applications. It is an extension of the Spring Framework, offering simplicity, speed, and ease of use.

MYSQL

MySQL is a reliable and widely used open-source relational database management system (RDBMS). It is designed to efficiently store, retrieve, and manage structured data.

Integration

Angular serves as the front-end layer, offering an intuitive and interactive interface for faculty and students. **Spring Boot** acts as the middle layer, implementing business logic, handling requests, and managing interactions with the database. **MySQL** forms the back-end layer, storing and retrieving data securely.

5 . SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS

DEVELOPMENT ENVIRONMENT

- Processor: Quad-core CPU (Intel Core i5/i7 or AMD Ryzen 5/7).
- RAM: Minimum 8 GB (16 GB recommended for optimal performance).
- Storage: SSD with at least 256 GB of available space.

SOFTWARE REQUIREMENTS

DEVELOPMENT TOOLS

1. FRONTEND DEVELOPMENT:

- **Framework:** React.js (latest stable version).
- **Node.js:** v16.x or later (required for React development and npm).
- **Package Manager:** npm (bundled with Node.js) or Yarn for dependency management.
- **Design Libraries:** Bootstrap, Material-UI for responsive UI components.

2. BACKEND DEVELOPMENT:

- **Framework:** Node.js with Express.js for API development.
- **Integrated Development Environment (IDE):** VS Code with Node.js extensions.
- **Runtime Environment:** Node.js LTS version.

3. AI AND DATA SCIENCE TOOLS:

- **Libraries:** TensorFlow, Scikit-learn, Pandas, NumPy for AI and data analysis.
- **Analytics Tool:** Power BI or Tableau for data visualization.

4. DATABASE:

- **Database Management System:** Firebase for real-time data handling or MySQL for relational data.
- **Database Interface Tools:** MySQL Workbench (optional) for visualization and queries.

OPERATING SYSTEM

Windows 10/11, macOS Monterey or later, or a Linux distribution (e.g., Ubuntu 20.04 LTS).

6 . CONCLUSION

The proposed web-based platform redefines the influencer marketing landscape by providing an innovative, transparent, and efficient solution for connecting companies with influencers. Through its AI-driven matchmaking system, the platform overcomes the limitations of traditional collaboration methods, such as inefficiencies, lack of transparency, and accountability issues. By incorporating features like secure communication, automated contract management, and real-time performance analytics, the system ensures seamless and reliable operations for all stakeholders.

The integration of modern technologies, including React.js for a dynamic front-end, Node.js for efficient back-end processing, and Firebase for real-time data management, has resulted in a robust and scalable application. These technologies collectively enable high performance and adaptability to future requirements, making the platform suitable for a wide range of industries and campaign scales.

By addressing critical gaps in influencer marketing, this platform fosters trust, enhances accountability, and streamlines collaboration processes. As the digital marketing landscape continues to evolve, this system sets a benchmark for creating transparent, data-driven, and user-friendly solutions in the field.